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Review article

Ergonomics in violin and piano playing: A systematic review

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ABSTRACT

This systematic review aimed to evaluate whether muscle activity and playing-related musculoskeletal disorders are associated with musicians' anthropometrics and their instrument size or set-up during violin and piano performance. Studies were retrieved systematically from six databases on 1 April 2019 combined with hand searching results. The Appraisal tool for Cross-Sectional Studies (AXIS tool) was used to evaluate the methodological quality of the included papers. A total of twenty articles were identified. Most included studies focussed on either the adjustment of the shoulder rest in violinists, or the hand size in pianists. However, methodological quality was inconsistent. The electromyography data reported by the included studies were not appropriately processed and interpreted. Studies generally reported the use of a shoulder rest changes muscle activity and smaller hand size is correlated to increased playing-related musculoskeletal disorders incidence. However, no conclusions can be drawn due to heterogeneity and low quality of methodology in the available literature.

1. Introduction

The workload of professional musicians was reported to be moderate to high during concerts and their practices and performances often involve highly repetitive and complex movements for prolonged periods of time, making them highly susceptible to physical problems (Iñesta et al., 2008; Quarrier, 1993). Reports of high occurrences of musculoskeletal complaints in professional musical instrumentalists are widespread and have been increasingly studied over the past few decades. A recent systematic review showed the point prevalence of musculoskeletal complaints in professional musicians could reach up to 68% and the lifetime prevalence ranged between 62 and 93% (Kok et al., 2016). Even amateur musicians reported a similar rate of playing-related musculoskeletal disorders (PRMDs) in an investigation of one-year prevalence (Kok et al., 2018). String and keyboard instruments, violin and piano particularly, are the most common musical instruments being played by students learning instrumental music (Ranelli et al., 2011) and the reported prevalence of PRMDs has been reported to be higher than in the general population of professional musicians (Amaral Correa et al., 2018; Kochem and Silva, 2018). In terms of the body regions affected by PRMDs, the upper limbs and neck are the area most commonly reported to be affected in both instrument types (Amaral Correa et al., 2018; Kochem and Silva, 2018). With these alarmingly high occurrences of injury and pain rates in mind, it is important to understand how to reduce the risk of incurring PRMDs during violin and piano playing.

The compatibility of the instrument with human physical abilities and characteristics is one of the principles in ergonomics (Corlett and Clark, 1995). Inadequate instrument set-ups and sizes that mismatch the player's physical attributes is frequently cited as one of the risk factors contributing to the development of PRMDs in violin and piano performers (Ackermann and Adams, 2003; Boyle, 2012; Shields and Dockrell, 2000). For violinists, playing in a sustained asymmetric posture is inevitable and it has been observed that the left upper limb, which is the side that supports the instrument, is predominantly symptomatic in upper string players (violinists and violists) (Kochem and Silva, 2018; Ramella et al., 2014). The use of adaptive accessories, such as a chin rest and/or a shoulder rest, has been proposed to be a way to improve the interface between the instrument and the player (Castleman, 2002; Okner et al., 1997). For pianists, a "one-size-fits-all" keyboard has been the available industry standard with respect to the size of the keyboard in modern piano manufacturing, teaching and performing (Booker and Boyle, 2011; Donison, 2000). However, recent demographic evidence proposes that this standardised piano keyboard discriminates against many pianists, especially the female players, who have relatively smaller hand sizes (Boyle et al., 2015). Controversies persist in best methods to ergonomically fit the instrument to the player among players and pedagogues of both these instruments, and no reviews to date have been conducted to determine whether a collection of evidence to support ergonomic ideals exists (Booker and Boyle, 2011;

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Abbreviations

EMG	electromyography
sEMG	surface electromyography
PRMDs	playing-related musculoskeletal disorders
PICOS	participant, intervention, control, outcome, study design

Homfray, 2007; Lacraru, 2014; Thomsen, 2011).

Electromyography (EMG) has been a useful tool to directly measure muscle activity levels and patterns, and may assist in evaluating risks of muscle overload during repetitive upper limb movements in work activities (Gazzoni et al., 2016). Recent biomechanical studies on violin and piano playing have employed EMG techniques to identify problems with playing techniques and provide information for preventing PRMDs (Kjelland, 2000; Schemmann et al., 2018; Trouli et al., 2013).

The objective of this systematic review is to evaluate, based on current evidence, whether musicians' (1) anthropometrics or (2) instrument size and/or set-up affect muscle activity and/or PRMDs during violin and piano performance.

2. Methods

2.1. Search strategies

The systematic review search was conducted on 1 April 2019 in the following databases: MEDLINE (1946–present), EMBASE (1974–present), SCOPUS (1970–present), ERIC (1966–present), CINAHL (1982–present) and Music Index (1973–present). The combination of keywords used in systematic literature search were discussed and selected by the authors in this paper. The resulting search terms were grouped as outlined in Table 1. Hand-searching, personal collections and exploding references were also used to augment the search results.

2.2. Study selection

The eligibility criteria for the studies were listed following the PICOS (participant, intervention, control, outcome, study design) model to reflect the research question in Table 2. Non-English-language papers were included in the review. Articles were initially screened by examining the titles and abstracts, followed by full text evaluation for their eligibility according to the inclusion criteria.

Table 1
Search strings.

("occupational" OR "pain" OR "injur*" OR "disorder*" OR "overuse*" OR "over-use*" OR "misuse*" OR "mis-use*" OR "repetitive" OR "WRULD*" OR "OOS*" OR "RSI*" OR "PRMD*" OR "PRP" OR "muscul*" OR "muscle" OR "electromyogr*" OR "electro-myogr*" OR "EMG" OR "biomechanic*")
AND (("music*" W/3 ("perform*" OR "instrument*")) OR "musician*" OR "violin*" OR "pian*")
AND ("bio-engineering" OR "bioengineering" OR "human engineering" OR "human factors" OR "ergonomic*" OR "equipment design" OR "redesign*" OR "re-design*" OR "chinrest" OR "chin rest" OR "shoulder rest" OR (("size*" OR "small*" OR "reduced") W/3 "keyboard*") OR "anthropometr*" OR "postur*" OR "height" OR "length" OR "span" OR "width" OR "breadth" OR "shape" OR "distance" OR "depth" OR "circumference")

"*" and "W/3" are search syntax used in the database search which may be different among the databases. "*" was used as truncation to find any extension of the root term; "W/3" functioned as word adjacency to find words or phrases within 3 words from one another in either order.

WRULD, work-related upper limb disorder; OOS, occupational overuse syndrome; RSI, repetitive stress injury; PRMD, performance-related musculoskeletal disorder; PRP, playing-related pain; EMG, electromyography.

Table 2
Eligibility criteria in PICOS model.

	Inclusion	Exclusion
Population	Violinists[a] and/or pianists within normal physical limit	Other instrumentalists or musician such as woodwind players, brassists, marching band musicians, vocalists, etc.
Intervention (exposure)	Different instrument set-ups (size of instrument or ergonomic modification on instrument) and/or anthropometric parameters	Studies not including ergonomic factors or anthropometric parameters in the data analysis and results
Comparator (applicable in some cross-sectional studies)	Musicians without PRMDs or other healthy controls	
Outcome	Instrument set-up and/or anthropometric difference in or association with PRMDs or related measures	No outcomes that relate anthropometrics or instrument set-up to PRMDs, or related measures
Study design	Any study design except case studies, qualitative research and review	

[a] Studies with target subjects consisted of violinists and violists were included.

2.3. Methodological quality assessment and data extraction

The methodological quality of all included studies was assessed using the Appraisal tool for Cross-Sectional Studies (AXIS tool) (Downes et al., 2016). The AXIS tool consisting 20 questions which mainly focus on the presented methods and results. Seven questions (1, 4, 10, 11, 12, 16 and 18) related to quality of reporting, seven (2, 3, 5, 8, 17, 19 and 20) questions related to study design quality and six related to the possible introduction of biases in the study (6, 7, 9, 13, 14 and 15). Since the development of AXIS tool was based on the premise of a degree of a researcher's discretion, not all the questions can be weighted equally during the assessment of study quality. The authors of this study selected six (1, 8, 9, 11, 12, 15) of the 20 questions as the critical ones that need to be answered. The included studies which failed to fulfil any of these six questions were considered to have lower quality in their methodology. For those involving using EMG as outcomes, a checklist adapted from criteria by the International Society of Electrophysiology and Kinesiology (ISEK) was used to evaluate the quality of reporting (Table 4) EMG (Armijo-Olivo et al., 2007; Merletti, 1999). Twenty-six questions were listed in the checklist. Questions 1 to 7 scrutinise the reports on specifications and applications of electrodes. Questions 8 to 12 focus on the details of the use of detection and amplification. Questions 13 to 22 examine whether filter, signal processing and sampling information were provided. Questions 23 to 26 identify the provision of normalisation information. These questions were answered with either "yes," "no," or "unknown." For questions that are not applicable to evaluate, the questions were noted as "NA". Information extracted from the critically reviewed papers was listed and included study design, demographics of study population (e.g. type of musician, population size, gender, age, playing history, etc.), outcome measurements, results and conclusions.

2.4. Research involvement

All study selection, methodological quality assessment and data extraction were conducted by the first author (JC) and re-evaluated by the second author (MH). If consensus was not made in any part of the above manners, a meeting with the third author (BJA) would be arranged and disagreements would be resolved through discussion.

3. Results

3.1. Search results

A total of 2395 records were found from the search in databases. According to the inclusion and exclusion criteria listed in Table 2, 256 citations were screened on title and abstract after removing duplicates. Further abstract and/or full-text examination were conducted and finally 20 studies including the results of hand search (n = 4) were selected for methodological quality assessment and data extraction. A flowchart of the literature search and selection is presented in Fig. 1. In these 20 research articles, 7 were investigating violinists (two of the studies included violists as subjects) and 13 were investigating pianists.

The results of the methodological quality assessment and the quality of reporting EMG are presented in Tables 3 and 4. Most studies demonstrated clear aims and addressed the aims with appropriate designs. The general methodology and internal consistency were also adequate in most of the selected articles. However, the main methodological flaws found in the collected studies, according to the six critical questions selected by the authors, including unclear aims, inappropriate use of measurements, untrialled measurements, insufficient descriptions in overall methods, and inadequate basic data information. One study had an unclear aim which made the article difficult to be assessed in the critical appraisal process (Blum and Ritter, 1990). Four studies showed low measurement validity and reliability (Allsop and Ackland, 2010; Furuya et al., 2006; Rabuffetti et al., 2007; Wristen and Hallbeck, 2009). Three studies presented a significant lack of methodological information (inadequate description of outcome measurements and/or statistical methods) (Blum and Ritter, 1990; De Smet et al., 1998; Wristen and Hallbeck, 2009) and five papers did not provide sufficient basic data of the study participants (Blum and Ritter, 1990; Farias et al., 2002; Rabuffetti et al., 2007; Sakai et al., 2006; Wristen and Hallbeck, 2009). Five studies in total used surface electromyography (sEMG) as the main or complementary outcome. Three of them provided relatively adequate (fulfilled the majority of the criteria) EMG reports in methodology (e.g. information about the use of electrodes, filter and normalisation procedures, etc.) (Kok et al., 2019; Levy et al., 1992; Wong and Lei, 2015); the other two studies reported significantly insufficient EMG protocols (Rabuffetti et al., 2007; Wristen and Hallbeck, 2009).

3.2. Anthropometric measurements and instrument set-ups

The description of all included studies is presented in Table 5. In the 7 research articles on the violin, 3 scrutinised the influence of anthropometric parameters and yet used different outcome measures (e.g. pain problem severity, EMG results from changing the shoulder rest and force and pressure on the chin rest) and the results were heterogeneous (Ackermann and Adams, 2003; Levy et al., 1992; Okner et al., 1997). Four studies investigating the effect of shoulder rest on muscle activities involved using sEMG. Although all of these four studies suggested the adjustment of shoulder rest had different degrees of changes to the muscle activities, two did not either conduct EMG normalisation or

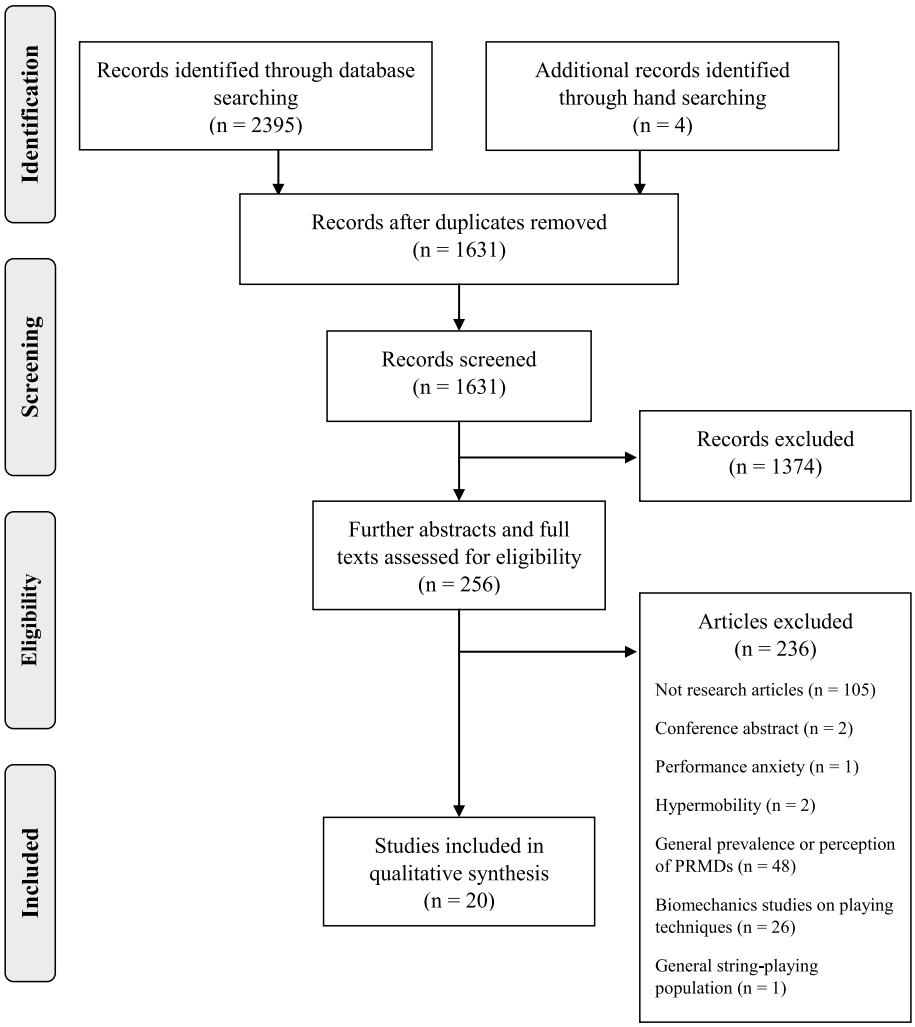

Fig. 1. PRISMA flow diagram of literature screening process.

Table 3
Results of methodological quality assessment.

group	Study	Questions																				Total (items)
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
violin	Ackermann & Adams (2003)	Y	Y	N	Y	Y	N	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	N	U	Y	14
	Blum and Ritter (1990)	N	Y	N	Y	Y	N	N	Y	Y	N	N	N	Y	NA	Y	Y	Y	N	U	N	9
	Kok et al. (2019)	Y	Y	N	Y	Y	Y	N	Y	Y	N	Y	Y	Y	NA	Y	Y	Y	Y	U	Y	15
	Levy et al. (1992)	Y	Y	N	Y	Y	N	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	U	N	14
	Okner et al. (1997)	Y	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	U	Y	16
	Rabuffetti et al. (2007)	Y	Y	N	Y	N	Y	N	N	Y	Y	Y	N	Y	NA	Y	Y	Y	Y	U	N	12
	Wong and Lei (2015)	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	N	Y	Y	N	14
piano	Allsop and Ackland (2010)	Y	Y	N	Y	Y	N	N	Y	N	Y	Y	Y	N	NA	Y	Y	Y	Y	U	Y	13
	Bruno et al. (2008)	Y	Y	N	Y	Y	N	N	Y	Y	Y	Y	Y	N	NA	Y	Y	Y	N	U	Y	13
	De Smet et al. (1998)	Y	Y	N	N	N	N	N	Y	Y	Y	N	Y	Y	NA	Y	Y	Y	N	U	N	10
	Farias et al. (2002)	Y	Y	N	Y	Y	N	N	Y	Y	N	Y	N	Y	NA	Y	Y	N	N	U	N	10
	Furuya et al. (2006)	Y	Y	N	Y	Y	N	N	Y	N	Y	Y	Y	N	NA	Y	Y	Y	Y	U	N	12
	Kaufman-Cohen et al. (2018)	Y	Y	N	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	Y	Y	17
	Lai et al. (2015)	Y	Y	N	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	Y	Y	17
	Sakai et al. (2006)	Y	Y	N	Y	N	N	N	Y	Y	Y	Y	N	Y	NA	Y	Y	Y	Y	Y	N	13
	Sakai and Shimawaki (2010)	Y	Y	N	Y	N	N	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	Y	Y	15
	Wristen and Hallbeck (2009)	Y	Y	N	Y	N	Y	N	N	Y	Y	N	N	Y	NA	Y	N	Y	Y	Y	Y	12
	Yoshimura et al. (2006)	Y	Y	N	Y	Y	N	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	U	Y	15
	Yoshimura et al. (2008)	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	N	U	Y	16
	Yoshimura and Chesky (2009)	Y	Y	N	Y	Y	N	N	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	U	Y	15
No. of studies meeting each criterion item		19	20	2	19	14	6	1	18	18	17	17	15	17	0	20	19	18	14	6	12	

Notes: Yes (Y), No (N), unknown (U), not applicable (NA).
1. Were the aims/objectives of the study clear?.
2. Was the study design appropriate for the stated aim(s)?.
3. Was the sample size justified?.
4. Was the target/reference population clearly defined?.
5. Was the sample frame taken from an appropriate population base?.
6. Was the selection process likely to select subjects/participants that were representative of the target/reference population under investigation?.
7. Were measures undertaken to address and categorise non-responders?.
8. Were the risk factor and outcome variables measured appropriate to the aims of the study?.
9. Were the risk factor and outcome variables measured correctly using instruments/measurements that had been trialled, piloted or published previously?.
10. Is it clear what was used to determined statistical significance and/or precision estimates?.
11. Were the methods sufficiently described to enable them to be repeated?.
12. Were the basic data adequately described?.
13. Does the response rate raise concerns about non-response bias?.
14. If appropriate, was information about non-responders described?.
15. Were the results internally consistent?.
16. Were the results presented for all the analyses described in the methods?.
17. Were the authors' discussions and conclusions justified by the results?.
18. Were the limitations of the study discussed?.
19. Were there any funding sources or conflicts of interest that may affect the authors' interpretation of the results?.
20. Was ethical approval or consent of participants attained?.

provide relevant information (Levy et al., 1992; Rabuffetti et al., 2007); one provided EMG normalisation information but did not report the EMG results (e.g. mean or peak values) (Wong and Lei, 2015) and only one detailed both (Kok et al., 2019). Only one study surveyed the prevalence of 'fiddler's neck' (a chronic dermatitis on the left side angle of the jaw) and treatments used from interviews suggesting over 50% success rate in treating fiddler's neck while changing the chin rest set-up. However their descriptive results were not compared with controls (Blum and Ritter, 1990).

In the 13 studies on the pianists, all research involved investigation of the size of hands in relation to piano performance variables. Ten articles used the history of playing-related pain and/or injury as the primary outcome and only 3 involved the results of using research tools such as sEMG and motion capture systems as outcome variables. The hand size parameters were also inconsistently measured across studies. Of the collected 13 pianist studies, 8 measured the distance between physical landmarks of the hand to record the size of hand (De Smet et al., 1998; Kaufman-Cohen et al., 2018; Lai et al., 2015; Sakai et al., 2006; Wristen and Hallbeck, 2009; Yoshimura and Chesky, 2009; Yoshimura et al., 2006, 2008). One study assessed the hand anthropometrics using x-rays (Sakai and Shimawaki, 2010) and one grouped the hand sizes by taking major tenth on piano keyboard as reference (Farias et al., 2002). Three studies collected the hand size data as reported or traced by the participants themselves (Allsop and Ackland, 2010; Bruno et al., 2008; Furuya et al., 2006). Regarding the correlation between the hand size and the prevalence of PRMDs, 10 studies indicated a smaller hand size was associated with a greater risk for playing-related pain or injuries (Bruno et al., 2008; De Smet et al., 1998; Farias et al., 2002; Kaufman-Cohen et al., 2018; Lai et al., 2015; Sakai et al., 2006; Sakai and Shimawaki, 2010; Yoshimura and Chesky, 2009; Yoshimura et al., 2006, 2008) and 2 suggested hand size did not significantly influence the occurrence of PRMDs (Allsop and Ackland, 2010; Furuya et al., 2006). By comparing the methodologies and primary outcomes, 6 (physically measuring the subject's hand size) (De Smet et al., 1998; Farias et al.,

Table 4
Checklist of the quality of reporting EMG.

Group	Study	Questions																										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	Total (items)
Violin	Kok et al. (2019)	Y	Y	Y	Y	N	Y	Y	Y	N	N	N	N	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	Y	NA	18
	Levy et al. (1992)	N	N	N	N	Y	N	Y	Y	Y	Y	N	N	Y	N	Y	Y	Y	NA	NA	Y	Y	Y	N	NA	NA	NA	12
	Rabuffetti et al. (2007)	N	N	N	N	N	N	N	Y	N	N	N	N	N	N	N	N	N	N	N	N	Y	N	N	N	N	N	2
	Wong and Lei (2015)	N	Y	N	N	Y	N	N	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	NA	NA	NA	16
Piano	Wristen and Hallbeck (2009)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	NA	0
No. of studies meeting each criterion item		1	2	1	1	2	1	2	4	2	2	1	0	3	2	3	3	3	2	0	2	4	3	2	1	1	0	

Notes: Yes (Y), No (N), unknown (U), not applicable (NA).
1. Electrode material.
2. Electrode geometry.
3. Electrode size.
4. Use of gel, paste.
5. Skin preparation.
6. Interelectrode distance.
7. Electrode location.
8. Type of amplification.
9. Input impedance.
10. Common mode rejection ratio.
11. Signal-to noise ratio (the ratio of the EMG signal during muscle contraction versus the unwanted electrical signal recorded when the muscle is at rest, i.e. the baseline noise.).
12. Actual gain used.
13. Low and high pass filters.
14. Filter types.
15. Low- and/or high-pass cut-off frequencies.
16. Slopes of the cut-offs.
17. Rectification.
18. Smoothing.
19. Root-mean square time period over which the average was calculated.
20. Integrated EMG.
21. Sample rate.
22. Number of bits, model manufacturer of A/D card used to sample data into the computer.
23. Method used for normalisation procedure.
24. Training procedure to obtain MVC when applicable.
25. Position used.
26. Velocity/elongation information when needed.

Table 5
Descriptions of included studies.

Group	Author	No. of participants (M:F)	Age range (mean ± SD) (years)	Playing history	Instrument set-up	Anthropometric measures	Injury, pain and related measures (outcome variables)	Results
Violin	Ackermann and Adams (2003)	30 violinists and 2 violists (9:23)	19-60 (27.0)	Years of playing (mean): at least 10 (20.0) years Daily practice time (mean): 1–6 (3.5) hours	No instrument set-up information	Hand span - Tip of thumb to tip of 5th finger - Tip of 2nd to tip of 5th finger Little finger length Web space between 1st and 3rd metacarpal ROM and length of bilateral upper limbs ROM and length of cervical spine	PRMDs history questions - Pain site - Current pain by using numerical pain scale (0-to-3) - Previous pain episodes	Increasing right arm pain severity associated with larger right 2nd-5th hand span and shorter right lower arm ($R^2 = 0.31$, $p = 0.005$). Pain severity in the thoracic spine increased with shorter right lower arm and the increased degree of left cervical lateral flexion on the violin ($R^2 = 0.36$, $p = 0.002$). Shorter the front of the neck length significantly predicted higher lower back pain severity ($R^2 = 0.16$, $p = 0.022$).
Violin	Blum and Ritter (1990)	523 professional upper string players (411: 112)	41	Information not provided	Different treatments (including changing the set-up of chin rest) used for 'fiddler's neck' was collected	No anthropometric measure made	The prevalence of 'fiddler's neck' ('practice mark') was collected during interview.	62% of these musicians had a 'practice mark' in different forms. In the 241 players with fiddler's neck seeking treatments, 63 were treated with modified chin rests or new chin rests with 57% success rate. 32 used padded chin rest and almost 80% of them had positive results. 2 received chin rests with plaster and both had successful responses.
Violin	Kok et al. (2019)	20 professional violinists (4:16)	21-38 (29.4 ± 3.7)	Average age started violin: 6.1 ± 1.5 years Average weekly practice time: 28.1 ± 13.6 h	Participants used their own shoulder rest and set in 4 conditions: - No shoulder rest - Lowest position (left feet 3.0 cm, right feet 2.0 cm) - Middle position (left feet 5.0 cm, right feet 4.0 cm) - Highest position (left feet 7.0 cm, right feet 6.0 cm).	No anthropometric measure made	Force (N) measured by using an instrumented chin-rest with a built-in force sensor Surface EMG measuring bilateral SCMs, bilateral upper TRs and left AD; signals in each muscle normalised to its MVIC VAS scoring from 0 (comfortable as imagine) to 10 (unplayable) to measure violinist's comfort level.	Higher shoulder rest condition associated with decreased subjective playing comfort (increased VAS score) ($p < 0.001$). Higher shoulder rests were related to a higher violin fixation force ($p < 0.001$). An elevated in shoulder rest height had a significant effect on increasing the activity of the left anterior deltoid ($p = 0.025$) but showed no impact on SCMs and upper TRs.
Violin	Levy et al. (1992)	15 experienced violinists (7: 8)	17-36 (26.2)	Average years of playing: 19.1 years; Average daily practice time: 4.5 h	A Kun model of shoulder rest was used to create 2 set-up conditions: - Use of a shoulder rest	Neck-shoulder dimension parameters - Seventh cervical vertebra (C7) to the inion (C7IN)	Surface EMG signals of left BI, AD, TR and right SCM was collected. Normalisation of the muscle activity	Reduced EMG in the left TR ($p = 0.030$) and the left SCM ($p = 0.004$); increased EMG in the AD ($p = 0.008$) and no significant change

(*continued on next page*)

Table 5 (*continued*)

Group	Author	No. of participants (M:F)	Age range (mean ± SD) (years)	Playing history	Instrument set-up	Anthropometric measures	Injury, pain and related measures (outcome variables)	Results
					- No use of a shoulder rest	- Mastoid to C7 (MSC7) - C7 to acromion (C7AC) - Mastoid to acromion (MSAC) - Midclavicle to mastoid (MCLM) - Acromion to midclavicle (AMCL)	signals was not reported.	in the BI when the shoulder rest was used. MSC7 positively predicted the reduced muscle activity as using a shoulder rest in TR and right SCM while playing at first position ($R^2 > 0.48$, $p < 0.025$). The combinations of MSC7 and C7AC as well as MCLM and AMCL were positively correlated with the decreases of TR and right SCM muscle activity ($R^2 > 0.42$, $p < 0.05$) while shifting with a shoulder rest.
Violin	Okner et al. (1997)	10 professional violinists (1:9)	26-49 (36.0 ± 7.7)	Years of playing (mean ± SD): 17-38 (27.5 ± 6.9) years; Weekly practice time (mean ± SD): 3–50 (17.0 ± 11.9) hours	7 instrument set-up conditions: 6 kinds of combinations with 3 types of chin rests (Wolf Maestro, Guarneri and Cliff Johnson) and 2 shoulder rest models (Resonans and Playonair Deluxe) and subject's own set-up	Neck length (mastoid to medial end of clavicle) range (mean): 17.3–21.4 cm (19.3 ± 1.4 cm) Shoulder width (C7 to acromion) range (mean): 17.9–21.9 cm (19.9 ± 1.2 cm)	Force and pressure parameters (measured via a sensor mat over the chin rest): - Peak Pressure (N/cm2) - Peak Force (N) - Pressure–Time Integral (N·sec/cm2) - Force–Time Integral (N·sec) - Total contact area (cm2)	Wolf Maestro chin rest significantly produced less pressure and force than other chin rests and shoulder rests ($p < 0.01$). Playonair shoulder rest generated less peak pressure and total contact area compared to Resonans ($p < 0.05$). The neck length were not significant predictors of any variance in pressure or force. The Shoulder width was positively significant only for pressure parameters when playing the selected Bruch concerto ($p < 0.05$). The height of subject's own instrument set-up was positively correlated to the pressure and force readings ($p < 0.001$) primarily in the Handel composition.
Violin	Rabuffetti et al. (2007)	15 healthy, skilled violinists (7:8)	23-59 (40.9 ± 10.2)	Information not provided	A Kun model of shoulder rest was used to create 3 set-up conditions: - No shoulder rest - Lowest feet height (on average of 26.2 mm) - Highest feet height (on average of 38.9 mm)	No anthropometric measure made	Kinematics of head, upper limbs, the violin body bow measured by using optical motion capture system Surface EMG was applied to record the muscle activities of bilateral TRs and SCMs, right MD, left AD and extensors of the head. Signal processing and normalisation of	The increase in rest height was related to the change of playing movement, posture and instrument position ($p < 0.05$). The left SCM was the only muscle to show an activity significantly increasing with the height of the shoulder rest ($p < 0.05$).

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Table 5 (*continued*)

Group	Author	No. of participants (M:F)	Age range (mean ± SD) (years)	Playing history	Instrument set-up	Anthropometric measures	Injury, pain and related measures (outcome variables)	Results
							muscle activities were not reported.	
Violin	Wong and Lei (2015)	8 healthy violinists (3:5)	20–25	Average years of playing: 10.4 ± 3.8 years; Average weekly practice time: 5.6 ± 2.4 h	12 different shoulder rest height combinations (Kun model): - 3 different height levels on Feet A (on the shoulder side) - 4 different height levels on Feet B (on the chest side)	No anthropometric measure made	Surface EMG applied to record the muscle activities of bilateral upper TRs (EMG signals were normalised to reference EMG levels by asking participants holding a given load.	The main effect of raising the shoulder rest feet height of Feet A increased the muscle activity level of left upper TR as Feet B was kept in the same height ($p < 0.05$). Raising the height of Feet B with a same height of Feet A increased right upper TR muscle activity ($p < 0.05$). However, when Feet B was at the highest level, no significant effect on Left upper TR was found. The muscle activity of right upper TR dropped with the decreasing Feet A as the highest level of Feet B was sustained ($p < 0.05$).
Piano	Allsop and Ackland (2010)	505 pianists (154:351)	12–89	Years of playing: over 2 years	No instrument set-up information	Hand span Respondents measure their maximal hand span (thumb to 5th digit)	PRMDs history questions: - Experience of pain (yes/no) - Pain site - PRMDs during playing - Pain duration, intensity and quality Pain management and result	Response rate: 87.1%; ANOVA yielded no main effect of hand span on experience of PRMDs ($F [1, 504] = 2.6$, $p > 0.05$).
Piano	Bruno et al. (2008)	195 piano students (78: 117)	10.9–26.1 (16.7 ± 3.3)	Average years of playing: 8.9 years Average age started piano: 7.8 ± 2.2 years; Weekly practice time (mean): 6–40 (14.2) hours	No instrument set-up information	Hand size was measured by evaluating the outline of their dominant hand traced by the respondents.	PRMDs history questions - Symptoms interferes playing in the last 4 weeks - Pain site - The quality and the quantity of the symptoms	Hands sizes were significantly smaller in the affected piano students with only upper limb pain compared to those unaffected both in the young and in the adult group ($p < 0.05$).
Piano	De Smet et al. (1998)	66 pianists (33:33) 66 controls (not musicians) (33:33)	Study group: 18–32 (22.6) Control group: 18–32 (24.3)	Study group: Age started piano (mean): - Male: 5–18 (8.3) years - Female: 5–13 (8.2) years Average daily practice time: 3.4 h	No instrument set-ups and/or sub-grouping made	Hand span - Thumb to 2nd finger - Thumb to 5th finger Hand length Middle finger length Joint hypermobility	A questionnaire concerning musculoskeletal problems was administrated but details were not provided	The distance between the thumb and 2nd finger was significantly greater in male pianists without problems than in those with lesions. Females without injuries had a longer hand than those with injuries.
Piano	Farias et al. (2002)	222 pianists with repetitive strain injury (RSI) (71:151)	8–70	Information not provided	No instrument set-ups and/or sub-grouping made	Hand span - Thumb to little finger (the distance was measured by taking Major tenth (22 cm) as a reference) Morphotype of the hands: - Morphotype A: palm width is	Repetitive strain injury (RSI)	In 222 affected pianists, 133 (60%) were with small-sized hands and most of them were female (74.2%). 149 (67% females and 33% males) affected pianists were classified with hands of morphotype B

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Table 5 (*continued*)

Group	Author	No. of participants (M:F)	Age range (mean ± SD) (years)	Playing history	Instrument set-up	Anthropometric measures	Injury, pain and related measures (outcome variables)	Results
						similar to the length of palm. - Morphotype B: palm width is shorter than the length of palm		whereas 73 (68.1% females and 33.1% males) were morphotype A.
Piano	Furuya et al. (2006)	203 pianists consisted of high school students to senior players	15–60	Age started piano (mean ± SD): 1–11 (4.5 ± 1.6) years Years of playing (mean ± SD): 9–50 (18.7 ± 9.0) years Daily practice time (mean ± SD): 0.6–9.5 (2.7 ± 1.7) hours	No instrument set-up information	Hand span data was collected based on participants reporting their difficulties of reaching octaves on the questionnaire.	Rate of PRMD occurrences for each body area within the past 5 years	77% of the total participants suffered from PRMDs. 8% of the 203 respondents reported great difficulty, 11% had some difficulty, 53% reported little difficulty, and 28% had no difficulty. Chi-square tests revealed no significant effect of hand span on PRMD prevalence at any body portion.
Piano	Kaufman-Cohen et al. (2018)	15 piano students (9:6)	19-27 (21.7 ± 2.4)	Years of playing (mean ± SD): 6–20 (13.4 ± 3.5) years; Weekly practice time (mean ± SD): 10.5–39.0 (23.9 ± 8.1) hours	No instrument set-up information	Hand span - Tips of thumb to 5th finger Limb length Arm length Forearm length 3rd finger length Wrist flexion/ extension (assessed using 3D motion capture) Wrist radial/ulnar deviation (assessed using 3D motion capture) Elbow flexion/ extension (assessed using 3D motion capture)	Questions adopted from Standardised Nordic Questionnaires: - Number of symptomatic body parts in the past year - Number of symptomatic body parts in the past week - Number of symptomatic body parts interfering with ADLs in the past year	The hand span was correlated with the number of symptomatic hand joints ($r = -0.56 \sim -0.69$, $p < 0.05$). Maximal wrist extension and/or elbow flexion while playing the piano also correlated with the number of symptomatic musculoskeletal joints as controlling the hand span ($r = 0.53–0.67$, $p < 0.05$).
Piano	Lai et al. (2015)	20 pianist students (0:20)	(23.6 ± 6.34)	Average years of playing: 17.5 ± 4.7 years	No instrument set-up information	The pianists were divided into two groups based on their hand spans* (between the tip of thumb and small finger): Small-hand-span group (mean ± SD) - Left: 18.8 ± 0.3 cm - Right: 18.0 ± 0.2 cm Large-hand-span group (mean ± SD) - Left: 21.6 ± 0.5 cm - Right: 21.3 ± 0.6 cm *The cut-off for small and large hand spans were based on the distribution of the hand spans in a priori (small hand span (<18.2 cm) and large hand span (>20.2 cm)).	Force and pressure on the piano keys measured by using piezoresistive force sensors Hand motion captured by cameras with markers on joints in bilateral hands. Parameters including: - The ratio of maximal digit-to-digit abduction angle (RD-Dabd) - Range of motion (ROM) of finger and wrist flexion-extension (F/E) - Radial-ulnar deviation (R/U)	The RD-Dabd was significantly larger for the small hand-span pianists when playing both chords and octaves ($p < 0.05$). The ROM of wrist F/E was significantly larger for small hand-span pianists when playing chords ($p < 0.05$). There was no significant difference in the fingertip force between two groups.
Piano	Sakai et al. (2006)	10 pianists (4:6)	24-39 (29.0)	Information not provided	No instrument set-up information	The pianists were divided into two groups based on hand span (between	Hand motions captured by applying markers over the	Both the maximal and minimal abduction angles of the thumb were

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Table 5 (*continued*)

Group	Author	No. of participants (M:F)	Age range (mean ± SD) (years)	Playing history	Instrument set-up	Anthropometric measures	Injury, pain and related measures (outcome variables)	Results
						the tip of thumb and small finger)*: - Small-hand-span group (mean 20.0 ± 0.6 cm) - Large-hand-span group (mean 24.0 ± 1.3 cm) *The two groups were significantly different ($p < 0.05$)	dorsal side of the hand to measure: - Max./mini. abduction angle of the thumb - Max./mini abduction angle of the little finger - Range of motion (ROM) of the thumb and the little finger abduction	significantly larger in the smaller-hand pianists compared with the pianists with larger hand spans when playing octave ($p < 0.05$). When playing chord, the maximum and ROM of thumb abduction of small–hand-span pianists was significantly larger than that of large–hand-span pianists ($p < 0.05$). Only the ROM of the little finger abduction was significantly larger in the small–hand-span pianists than the large–hand-span pianists while playing chord.
Piano	Sakai and Shimawaki (2010)	220 pianists with overuse medical problem (34:186) 62 unaffected pianists (18:48)	Patient group: 18–65 (30.0) Unaffected group: 20–52 (33.0)	Information not provided	No instrument set-up information	Assessed using x-ray: -Abduction angle of the thumb -Abduction angle of the little finger - Hand span (tips of thumb to 5th finger) - Short thumb length (from MCP joint to the tip) -Long thumb length (from CP joint to the tip) - Little finger length - Middle finger length	Overuse medical problems were reported by the author examined	The patients with tenosynovitis, epicondylitis and muscle pain in the forearm and hand had smaller hand span and finger length than the controls. ($p < 0.05$)
Piano	Wristen and Hallbeck (2009)	24 small-handed pianists (sex was not indicated)	Not reported	10 novice-level pianists and 13 experts and 1 expert excluded due to noncompliance; 4 of the 24 participants reported previous playing-related injury	Two pianos with different sizes of keyboards: - 7/8-size (an octave measures- the total width of seven white keys: 14.1 cm) - Full-size (an octave measures- the total width of seven white keys: 16.5 cm)	All small-handed pianists: full-hand span from tip of fifth finger to thumb ≤8 inches (~20.3 cm)	Surface EMG was used to record the muscle activities of masseter, trapezius, the hand and finger flexors, and the hand and finger extensors. The descriptions of signal processing and normalisation were not reported.	No significant difference between 7/8- and full-size pianos in the muscle activities ($p > 0.05$).
Piano	Yoshimura et al. (2006)	35 piano major students (8:27)	21-41 (27.2 ± 5.0)	Age started piano (mean ± SD): 3–13 (6.04 ± 2.30) years; Weekly practice time (mean ± SD): 3–42 (24.83 ± 8.46) hours	No instrument set-up information	Assessed bilaterally: - Length of upper arm, forearm and hand length - Circumference of wrist - Index finger diameter - Hand volume - Hand span (did not specifically described the measurement)	Four pain questions were used as dependent variables rated from 0 to 10 by participants: - "Do you experience pain when playing?" - "Do you experience pain after playing?" - "Does pain stop you from playing the piano?"	Some anthropometric hand measurements were significant predictors and negatively correlated with each of the pain questions in the regression models ($R^2 = 0.40–0.60$, $p < 0.005$). The right 3–4 digit span was the most or second strongest factor.

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Table 5 (*continued*)

Group	Author	No. of participants (M:F)	Age range (mean ± SD) (years)	Playing history	Instrument set-up	Anthropometric measures	Injury, pain and related measures (outcome variables)	Results
						- Maximum. interval on keyboard - Maximum digit-to-digit span (angles)	- "How much of your playing is affected by your pain?"	
Piano	Yoshimura et al. (2008)	47 piano teachers (8:39)	15-75 (42.9 ± 16.4)	Age started piano (mean ± SD): 3-13 (6.2 ± 2.0) years Weekly practice time (mean ± SD): 0–28 (9.1 ± 8.0) hours; Weekly teaching time (mean ± SD): 1-40 (16.6 ± 9.6) hours;	No instrument set-up information	Assessed bilaterally: - Length of upper arm, forearm and hand length - Circumference of wrist - Index finger diameter - Hand volume - Hand span (did not specifically described the measurement) - Maximum. interval on keyboard - Maximum digit-to-digit span (angles)	Four pain questions were used as dependent variables rated from 0 to 10 by participants: - "Do you experience pain when playing?" - "Do you experience pain after playing?" - "Does pain stop you from playing the piano?" - "How much of your playing is affected by your pain?"	The piano teachers with less left-hand maximum interval on keyboard presented had more tendency to report experiencing pain when/after playing ($R^2 = 0.26–0.30$, $p < 0.05$).
Piano	Yoshimura and Chesky (2009)	35 piano major students (8:27) (same study participants in the previous paper (Yoshimura et al., 2006))	21-41 (27.2 ± 5.0)	Age started piano (mean ± SD): 3-13 (6.0 ± 2.3) years; Weekly practice time (mean ± SD): 3-42 (24.8 ± 8.5) hours	Two pianos with different sizes of keyboards: - 15/16-size (an octave measures- the total width of eight white keys: 174 mm) - Full-size (an octave measures- the total width of eight white keys: 188 mm)	Assessed bilaterally: - Length of upper arm, forearm and hand length - Circumference of wrist - Index finger diameter - Hand volume - Hand span (did not specifically described the measurement) - Maximum. interval on keyboard - Maximum digit-to-digit span (angles)	Pre- and post-performance pain as well as the level of pain and tension during playing were measured by using self-administrated 10-cm visual analogue scales	Regardless of the hand span, playing on the 15/16-size keyboard significantly reduced the playing-related symptoms compared to playing on the full-size ($p < 0.05$). The levels of pain and tension during playing were inversely correlated with the size of hand span on both 15/16- and full-size keyboards ($p < 0.05$). Participants with smaller hand spans (<212.4 mm) had more pain and tension than those with larger hand spans (>212.4 mm) ($p < 0.05$). The levels of pain were lower while playing on the 15/16-size keyboard compared to the full-size in the small-handed group ($p < 0.05$) but not in the large-handed group.

AD, anterior deltoid; BI, biceps brachii; CP, carpometacarpal; EMG, electromyography; MCP, metacarpophalangeal; MD, medical deltoid; MVIC, maximal voluntary isometric contraction; N, newton; PRMD, playing-related musculoskeletal disorder; ROM, range of motion; SCM, sternocleidomastoid; TR, trapezius; VAS, visual analogue scale.

2002; Kaufman-Cohen et al., 2018; Lai et al., 2015; Sakai et al., 2006; Yoshimura and Chesky, 2009; Yoshimura et al., 2006, 2008) plus 1 (measuring the bony ends on x-rays) (Sakai and Shimawaki, 2010) studies involving hand size measurements carried out by the researcher and pain/injury questions/history as primary outcome reported (or implied) pianists with smaller hands were subjected to higher rate/-intensity of PRMDs. In contrast, in the 3 papers adopting self-reported methods on hand size, 2 suggested that hand size did not significantly influence the occurrence of PRMDs (Allsop and Ackland, 2010; Furuya et al., 2006) and one indicated hand size was significantly smaller in the affected pianists (Bruno et al., 2008). The 2 motion capture studies (Lai et al., 2015; Sakai et al., 2006) concluded similarly to the 7 papers investigating participants' history of PRMDs with the hand size measured by researchers, that awareness should be raised as having

small hands was a risk factor for PRMDs since players tended to abduct their fingers more when playing chords and octaves. Yet one study using sEMG equipment to investigate the effect of narrower keyboard in a group of pianists with small hands did not find significant differences in muscle activities while comparing them to the results on a standardised keyboard (Wristen and Hallbeck, 2009). As the outcome measurement of hand size varied and inappropriate outcome reported in some studies — for example, the study by Farias et al. (2002) concluded their observation of more repetitive strain injuries in small-handed pianists and female players without having a control group — conclusions were not able to be drawn even with the same methodological quality.

4. Discussion

This systematic review sought to extract and analyse data relating to anthropometric and biomechanical research in relation to the musical instrument in violin and piano players. Two key results were generally found from the included studies: (1) the set-up of violin, the use of a shoulder rest, may change the muscle activity and other biomechanical measurements related to PRMDs and (2) pianists with smaller hand size present higher PRMDs incidence. However, these key findings were strongly limited by significant variation in both the physical characteristics measured and the instrument set-ups which were being investigated, leading to inconclusive results.

4.1. Lack of scientific research on violin set-up

Only a limited number of scientific studies were found on this topic, despite the frequent literature assertions of the importance of a good 'fit' between a player and their instrument in pedagogical and health literature. Insufficient methodological quality was a common feature of these studies (Tables 3 and 4). It is interesting that 4 of the 7 studies on violin set-up investigated the effect of using a shoulder rest by applying EMG, reflecting the long-term debates on this issue in the violinist community (Homfray, 2007). Only one research article was found that investigated the physical response to different combination of the chin rest and shoulder rest (Okner et al., 1997), despite the many opinions about this topic in violin pedagogy (Denig, 2017; Dinwiddie, 2007; Roberts, 2011). Clearly, far greater research is needed on the physical impact of ergonomic devices on the player with consideration of their anthropometric characteristics, using appropriate measuring devices and analytical approaches to increase methodological quality.

4.2. The size of hand as a potential risk factor for PRMDs in pianists

Most included studies (10 out of 13) indicated small-handed pianists may have higher risk of PRMDs and face certain degree of physical challenges because of their proposed anthropometrical limitation on the standardised keyboard size. Only two studies reported the size of hand was not correlated to the history of PRMDs and one showed no significant different of EMG in small-handed pianists while playing between full-size and 7/8-size pianos. To explain the non-significant findings in the studies by Allsop & Ackland and Furuya et al., it is plausibly caused by misreporting from subjective anthropometric measures. Some research studies on the validity of different anthropometric data collection methods of height and weight suggested the lack of accuracy in self-reported anthropometric measures (Danubio et al., 2008; Maukonen et al., 2018; Park et al., 2011; Tang et al., 2016; Yoon et al., 2014). Even though the other literature proposed self-reported height and weight were correlated as well as directly measured ones (Lipsky et al., 2019; Olfert et al., 2018; Ortiz-Panozo et al., 2017; Pasalich et al., 2014; Xie et al., 2014), it should be noted that anthropometric information reported by respondents can be subject to variables such as age, sex, sociodemographics, ethnicity, health status and other confounding factors (Danubio et al., 2008; Lipsky et al., 2019; Maukonen et al., 2018; Olfert et al., 2018; Ortiz-Panozo et al., 2017; Park et al., 2011; Shields et al., 2008; Tang et al., 2016; Yoon et al., 2014; Wright et al., 2015; Xie et al., 2014). The use of the direct (or supervised) measurement of anthropometrics should be preferred when collecting hand size data from pianists. However, it should also be noted in the included study by Bruno et al., the hand size was reported by measuring the self-traced outline of the hand by participants (Bruno et al., 2008), which was considered to be an objective approach to collect the hand anthropometric data. Despite the fact that 10 out of 13 included studies showed negative correlation between hand size and physical distress (Bruno et al., 2008; De Smet et al., 1998; Farias et al., 2002; Kaufman-Cohen et al., 2018; Lai et al., 2015; Sakai et al., 2006; Sakai and Shimawaki, 2010; Yoshimura and Chesky, 2009; Yoshimura et al., 2006, 2008), it should be borne in mind that the playing history of participants was poorly documented in some studies as it can confound the effect of hand size. The details of hand size measures across studies were also rarely specified and therefore it may be hard for the future research to repeat the methodology. The study by Wristen and Hallbeck was the only study that investigated the hand size issue by using EMG in pianists and reported non-significant findings. It is very likely that the poor reporting of EMG (Table 4) compromised the results.

4.3. Poor EMG signal processing and interpretation

There were only five studies that employed sEMG as an outcome measurement (Tables 4 and 5). However, the quality of reporting of EMG data in all of these included studies was questionable. One study described the use of sEMG without detailing how the signals were processed and the other was directly comparing the EMG values without normalisation (Levy et al., 1992; Rabuffetti et al., 2007). The EMG must be normalised to be able to compare EMG activity in the same muscle on different days or in different individuals or to compare EMG activity between muscles (Halaki and Ginn, 2012). Two studies inappropriately interpreted EMG results into level of force and comfort (Wong and Lei, 2015; Wristen and Hallbeck, 2009). Since the torque/EMG is not linear relationship and the lengths of muscle moment arms and motor strategy vary in each individual, predicting the force production from muscle activity is not a valid data conversion (Halaki and Ginn, 2012). Comfort/discomfort is usually measured by subject's objective and subjective responses (e.g. postural compensation and rating the severity of discomfort). Fatigue, in terms of physical performance, however, is a temporary condition induced by continued activity that can be quantified by using EMG (Lamond and Dawson, 1999; Tucker, 2003; Jarić et al., 1997).

4.4. Inconsistent measures of ergonomic variables across studies

It was evident that there is not a clear approach to measuring variables relating to violin and piano ergonomics and the performers themselves. As mentioned in Section 4.2, the measuring method of hand size varies across studies from self-reported to the hand span measured by the researchers. Even though for those directly measuring the hand anthropometrics, the details of how the hand size was collected were rarely specified. The investigated set-ups on the violin and piano were also different in the included studies (e.g. different combinations of the chinrest and shoulder rest, a range of shoulder rest setting conditions, different sizes of piano keyboard). A lack of clear description of research methodology and study population also raised concerns to the results in several selected literatures.

4.5. Limitations and recommendations for future research

A limitation of this study may be that the search strategy limited the papers retrieved relevant to the issue of musicians' physical anthropometrics and musical instrument set-ups. However, the search strategy was aimed to correlate this information with the impact of these variables on the outcomes of PRMDs or muscle activity effects. Another

limitation is that the heterogeneity of the study design and the questionable quality of methodology in the included articles precluded the ability to pool study data. Important information, such as the descriptions of study subjects and non-respondents, was also missing in most of the searched articles. Some studies did not use statistical analysis. Therefore, in this first attempt to gather research articles investigating the effect of instrument set-up and body measurement in the violin and piano performance, it is not possible to draw any conclusions. Future research should be conducted with careful study design, particularly with higher quality of methodology and better understanding in the measuring tools or protocols. For example, a consistent measurement of hand size by researcher in studies on piano performance and a clear reporting of EMG data by referring to ISEK and other well-written EMG literature. A detailed background information about study participants should be also emphasised in studies involving anthropometric measurements in violin and piano playing to assure that the study participants are representative to the target population.

5. Conclusions

The results of this systematic review found only scarce evidence available on instrument set-ups and player's anthropometrics and their impact on muscle activity and PRMDs in violin and piano performance. Studies generally reported the use of a shoulder rest changes muscle activity and smaller hand size is correlated to increased PRMD incidence. However, inconsistent methodological quality and measuring tools or protocols precluded the ability to pool data and form robust conclusions. Future research should be careful with study design and result reporting to enhance the quality of scientific evidence in performing art medicine.

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Declaration of competing interest

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